

Design elements affecting wafer temperature uniformity in multi-wafer production MBE systems

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ABSTRACT

In this paper we present the results of key experiments that have illuminated our understanding of the interactions between wafers, backing rings, platen and manipulator heater in a $7 \times 6''$ multi-wafer production MBE system. We present examples of detailed wafer temperature maps across platens of 6-in diameter GaAs wafers. Results are discussed in terms of a qualitative model that can guide the design of MBE system manipulators, platens and backing rings that will be capable of producing a high degree of temperature uniformity. One of the most important factors is that the backing ring must be of proper width to compensate for heat conducted into the wafer perimeter through contact with the hotter platen. The platen must also be far enough from the radiative heating source, and of low reflectivity, to prevent reflected power from causing localized temperature non-uniformities in the heater. Because of the importance of platen emissivity to the thermal environment of the wafers, users should be aware of unintended changes in emissivity as platens become coated through weeks or months of use.

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1. Introduction

Measurement and control of substrate temperature is critical to the reduction of process variation when manufacturing semiconductor epiwafers. It is well known that the temperature of the substrate during growth can have a significant influence on physical properties of the semiconductor epilayers, including electron and hole mobilities; minority carrier lifetimes; interface abruptness; surface roughness; incorporation, activation and diffusion of dopants; layer thicknesses; alloy compositions; and incorporation of undesirable impurities. Depending on the structures being grown, the distribution of important optical and electronic device parameters will often reflect the uniformity of wafer temperature during growth of the epiwafers to be processed. Therefore, substrate temperature reproducibility and uniformity is critical to achieving competitive electronic or optoelectronic device performance and yield.

In this paper we present examples of detailed temperature maps across platens of 6-in (150 mm) diameter, semi-insulating, double side polished GaAs wafers. The temperature maps were acquired using the k-Space BandiTTM [1], a commercially available system that infers temperature using the band-edge thermometry (BET) [2] method. We have published details elsewhere regarding

the development of two-dimensional temperature mapping capabilities using this system and the application of the system to automated control of manipulator temperature setpoint offsets [3]. In this paper we discuss the causes of various observed patterns in the wafer temperature maps and means for reducing temperature variation across the wafers. Temperature uniformity data presented in this paper was collected from Veeco Gen 2000 molecular beam epitaxy (MBE) systems; however, many of the general principles discussed should apply to any single or multi-wafer MBE systems where the wafers are being radiatively heated.

2. Experimental results and discussion

2.1. Manipulator power ratio tuning

Multi-zone manipulator heaters are commonly incorporated into single- and multi-wafer MBE systems to give users a means of tuning out non-uniformities that are radial with respect to the platen [4,5]. Temperature uniformity maps such as that seen in Fig. 1 can be used as an aid in determining the optimum power distribution between the multiple zones. This surface plot shows a non-optimized 8 °C temperature difference between the center of the center wafer and the peak temperature near the outer edge of the outer wafers. Using this data as a guide, and some additional iterations, we reduced this temperature difference to ~2 °C by lowering the outer zone heater power ratio from the initial value

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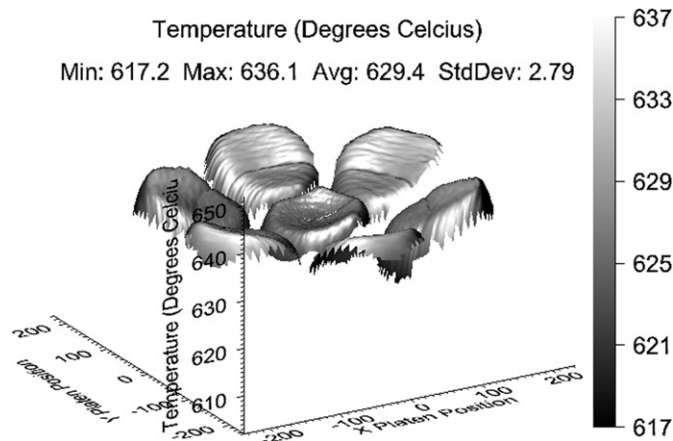


Fig. 1. Temperature map from a 7×6 " platen of GaAs wafers. Power supplied to the outer perimeter heat zone is non-optimal, causing a thermal gradient across the outer wafers.

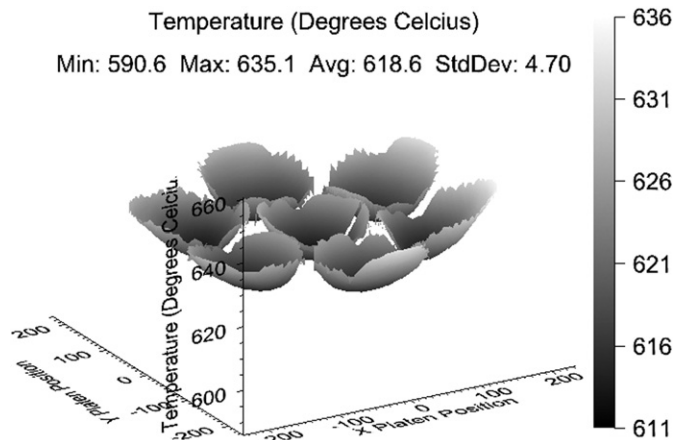


Fig. 2. Temperature map from a 7×6 " platen of GaAs wafers. A narrow backing ring and wide platen ledge contribute to excess heating at the wafer periphery.

of 8% to 4.3% of total applied manipulator heater power. Consequently, the calculated standard deviation of the measured data points dropped from 2.79 to 2.06 °C. Power ratio optimization using such temperature maps has become part of our standard MBE system setup procedure after any major maintenance of the growth chamber, cleaning of the platens or noticeable deviations in device performance variation across the wafers.

2.2. Backing ring and ledge width effects

Not all substrate temperature uniformity problems can be addressed with adjustments to manipulator heater zone power ratios. Note that Fig. 2 shows a temperature variation pattern with a strong component of radial symmetry with respect to wafer centers. This pattern is an indication that the inner diameter (I.D.) dimension of the wafer backing rings is not properly optimized for the I.D. of the ledge that each wafer rests upon.

Fig. 3 shows a schematic drawing of the cross-section of a platen pocket with a wafer resting on the circular ledge within it. The vertical arrows represent radiative heating from the manipulator heater. The wavy arrows represent heat conduction through the platen and into the periphery of the wafer through contact with the platen ledge. Because a graphite or molybdenum platen is more absorbing than a GaAs wafer, the platen is typically

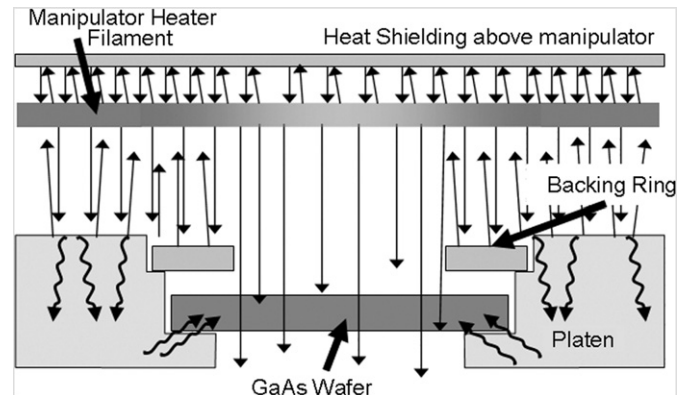


Fig. 3. Cross-sectional view of a platen pocket with a GaAs wafer resting on the ledge and a wafer backing ring in place above the wafer. Above the platen is the manipulator heater assembly.

hotter than the wafer once the system has stabilized at growth temperature. Therefore, there will be a positive net heat flow from platen to wafer through contact with the ledge.

Note the wafer backing ring in Fig. 3 which rests on an upper ledge within the platen pocket and is not in contact with the GaAs wafer. This backing ring acts as a heat shield, reflecting a portion of the heat radiating from the heater filament that would otherwise contribute to additional radiative heating of the wafer periphery. A properly designed wafer backing ring will overlap the wafer sufficiently to exactly compensate for heat flowing into the wafer through the ledge. Platens with a smaller ledge I.D. (wide ledge) will have increased thermal conduction from platen to wafer, and consequently require backing rings with a smaller I.D. (wide ring) for minimum temperature variation from wafer center to wafer periphery.

The example data shown in Fig. 2 is from a platen with a ledge I.D. of 144 mm and a backing ring I.D. of 144 mm. The elevated temperature at the periphery of each wafer indicates that the backing ring does not sufficiently compensate for heat conducting into the wafers through the relatively wide platen ledge. A smaller backing ring I.D. of 142 mm improves the temperature uniformity, slightly inverting the center to edge temperature gradient. A dimension between 142 and 143 mm is likely optimal for this ledge width.

Given that an optimum backing ring width exists for any ledge width, other factors should be considered when choosing a ledge width in the platen design. For example, a wide ledge might be needed merely to allow reproducible epilayer thickness measurements using a profilometer at the epitaxy exclusion step. Otherwise, a narrow ledge can potentially increase the number of useful devices available from a given wafer diameter. Surprisingly, one gains nearly 9% more epitaxial area on a 4-in wafer when the ledge overlap is reduced from 3 to 1 mm.

Aside from concerns with temperature reproducibility and uniformity under steady-state conditions, one may also be concerned with the maximum rate at which the wafers in the platen can be heated and cooled. For example, a more thermally agile system would allow for a shorter growth pause if changing substrate temperature between growth of an InGaAs layer and an AlGaAs layer. To test the effect of ledge I.D. on temperature ramp rate, we designed a platen that implemented three different ledge I.D. dimensions distributed between the six outer wafer pockets of an otherwise standard 7×6 " platen. Appropriate ring widths were selected for each wafer such that all maintained reasonably good temperature uniformity under steady-state conditions. Measurements of wafer temperature during an aggressive manipulator temperature ramp from a wafer temperature of 200–630 °C show

that the ledge width has a strong effect on the thermal agility of the wafers. Near the end of the setpoint ramp, the wafers on ledges of 148 mm I.D. were temporarily 70 °C hotter than the wafers on the 144 mm I.D. ledge. At steady state all the wafers were within a few degrees of each other.

Due to the high thermal capacitance of a 4 mm thickness molybdenum platen, even well-darkened platens will not heat as quickly as the wafers supported in the platens. A wider ledge increases coupling between wafer and platen, slowing the ramp rate of the wafer in the platen. During an aggressive ramp to increasing temperature, the center of the wafer will heat faster than the periphery, inducing a temperature gradient from center to edge. As this is enhanced by increased contact area with the platen, it may be that a combination of aggressive ramp rates and wider ledges will induce slip lines more readily than with platens that incorporate narrower ledges. We have not yet confirmed this experimentally, but we have confirmed that excessively wide backing rings, combined with a narrow ledge, will increase the production of slip lines in wafers. This is likely due to the large magnitude of the center-to-edge temperature gradient present at steady state.

If using graphite platens, the temperature gradient is reversed during aggressive ramping due to higher emissivity and lower volumetric thermal capacitance of graphite as compared to molybdenum. The platen increases in temperature faster than the semi-insulating GaAs wafers, and the periphery of the wafer becomes hotter than wafer center if the ring is properly dimensioned for good temperature uniformity at steady state.

2.3. Platen emissivity effects and influence of the platen-to-heater spacing

With near optimal backing ring width and manipulator power ratios, detailed temperature maps allowed us to discover patterns in the temperature maps that could not be affected substantially by any further adjustments to the parameters discussed above. The temperature map shown in Fig. 4 indicates a low-temperature valley in the center of the center wafer, in spite of the backing ring clearly being slightly too wide. (The temperature at the periphery of the center wafer has a downward trend in the last few millimeters approaching the backing ring.) Similarly baffling was the saddle-shaped temperature profile of the outer wafers.

After a number of experiments, we determined that both of these features are related to power being reflected back up into

the heater from annular regions of the platen that contain more molybdenum than GaAs under rotational averaging. As described by Johnson et al. [4] for single-wafer manipulators, the GaAs wafer is semi-transparent to radiation from the heater filament. This allows radiative cooling of regions of the heater that consistently remain above GaAs during platen rotation.

In the case of a 7 × 6" platen, two regions of the heater are most influenced by this localized radiative cooling. This includes the region of the heater that lies above the center wafer, and the annular region of the heater that lies above the portion of the platen where the molybdenum spacers between wafers in the outer ring are narrowest.

The non-uniformities can be reduced by both minimizing the reflected power from the molybdenum platen and by delocalizing the effects of this reflected power. Fig. 5 shows temperature profiles from 7 × 6" platens under various platen emissivity conditions. Each plot is a subset of data from a full platen map. Data points have been extracted along a single radial line starting at platen center and extending out along a line that bisects an outer wafer. Thus, the data point at radius 0.0 mm is the measured temperature at the center of the center wafer. With one exception, the applied manipulator heater power was nearly identical for each data set. The data set represented by solid diamond symbols was acquired from an upgraded manipulator heater design and did not use the same power settings.

A recently cleaned molybdenum platen is highly reflective and acts as a very effective heat shield between the manipulator heater filament and the growth chamber cryopanel. Therefore, reflected power is at a maximum prior to the first epitaxial deposition. The data represented by solid squares in Fig. 5 represents this condition, and the temperature variation across the center wafer exceeds 20 °C. Fortunately, this worst-case condition rapidly goes away within the first micron of GaAs deposition. The data represented by open triangles in the graph is taken from the same molybdenum platen after a few microns of GaAs deposition. This GaAs coating increases the emissivity of the lower face of the platen, reducing the platen's effectiveness as a heat shield between heater filament and cryopanel. The temperature variation across the center wafer is then reduced to < 11 °C, but is still unacceptable for many applications.

To increase the emissivity of the top side of the platen, the platen was loaded upside-down with molybdenum inserts and coated with ~0.5 μm GaAs. This two-side coated platen was then reloaded with GaAs wafers for acquisition of another wafer temperature map. The result is plotted using open circles. Use of

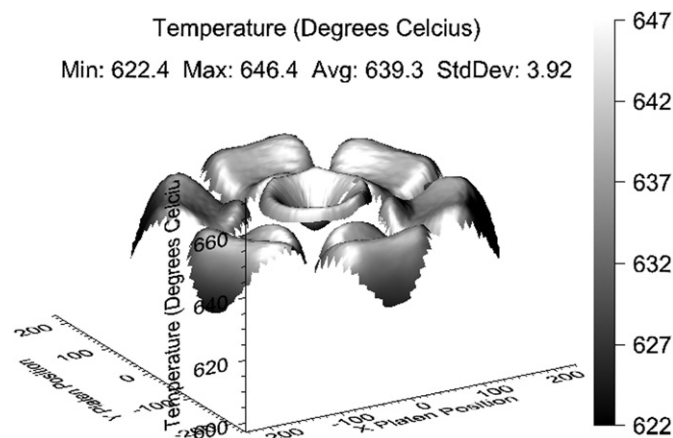


Fig. 4. Temperature map from a 7 × 6" platen of GaAs wafers. A reflective molybdenum platen and close spacing between heater and platen combine to cause a deep valley in the center wafer and a saddle-shaped profile in the outer wafers.

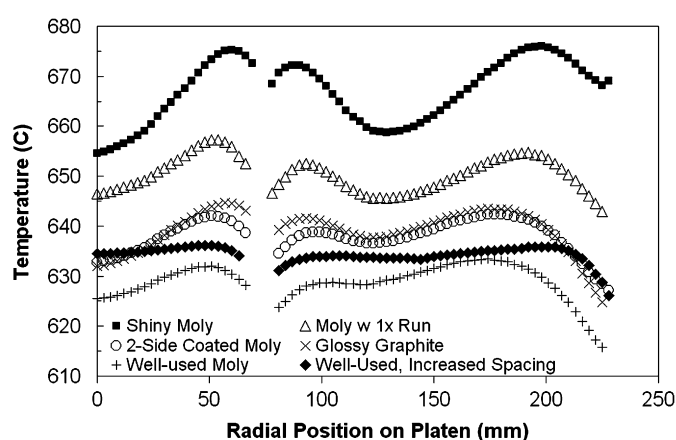


Fig. 5. Radial wafer temperature profiles from 7 × 6" platens under various platen emissivity conditions. Each data set represents temperatures measured along a radial line that extends outwards from platen center, bisecting a wafer in the outer ring.

a clean graphite platen with a glossy finish gave similar results, as plotted using 'x' symbols. In either case, the uniformity improvement is noticeable, but small. It is likely that a matte finish graphite platen would give further improvement.

The best uniformity, when using this relatively close platen-to-heater spacing, is plotted using the '+' symbols. This was obtained from a molybdenum platen that received no special treatment but many months of use in growing standard product epiwafers. We have observed that the top side of the molybdenum platen darkens with use, developing a coating that is often darker than the thick GaAs deposition found on the bottom side of the platen. We have not determined whether it is an oxide from repeated platen venting and heating in the preparation chamber, or if it is merely a thin coating of InGaAlAs from source material that makes its way into the space between platen and heater filament over time. Whatever the case, these results indicate that this condition is more effective than our intentional GaAs coating in terms of reducing wafer temperature non-uniformities. Between the initial, clean platen condition, and the final, well-used platen condition, the valley in the center wafer is reduced from a $>20^{\circ}\text{C}$ temperature range to a 6.5°C total range.

The most effective means of reducing the influence of platen emissivity on the temperature uniformity of the substrate heater is to delocalize the influence of the inevitable reflected power. This can be accomplished by increasing the distance between platen and heater. The data of Fig. 5 plotted in solid diamonds was acquired after upgrading the manipulator assembly with an approximately 0.75" increase to the heater-to-platen distance. This upgrade also necessitates a corresponding change to the heat shielding around the manipulator periphery. The data set shows that the temperature valley in the center wafer has been reduced to $<2^{\circ}\text{C}$, a great improvement on the result of 6.5°C when using a well-coated platen and the closer platen-to-heater spacing. Having conquered the dominant sources of temperature non-uniformity, we are currently working to reduce radiative losses at the outside periphery of the manipulator and the resulting temperature drop-off at the outer edge of the outer wafers.

3. Conclusions

We have acquired wafer temperature maps using a variety of platen and backing ring dimensions, platen-to-heater spacings and platen emissivity conditions. All of these factors can have a strong influence on the radial uniformity of temperature from center to edge of each wafer. A properly dimensioned backing ring offsets the effect of heat conducting into the wafer periphery through the ledge. Use of a narrower wafer ledge with corresponding optimal ring dimensions can allow faster wafer heating during an aggressive temperature ramp. Over the range of wafer ledge widths tested, we have found that the backing ring should

have an I.D. that is 1–2 mm smaller than the I.D. of the ledge. However, the higher absorption of single side polished or semi-conducting GaAs wafers may result in a larger ring I.D. being optimal when such wafers are in use with the same platen design. Temperature non-uniformities can also be induced by localized radiative cooling of the manipulator heater filament. The resulting wafer temperature variation can be reduced by increasing the distance between the heater and the platen, and by reducing the reflectivity of the platen.

The emissivity of the platen has a great deal of influence on wafer temperature uniformity, the manipulator setpoint required to arrive at a given wafer temperature target during growth recipes, and the rate at which wafers will follow a temperature setpoint ramp. Therefore, the gradually changing platen emissivity over months of production runs will effectively make "optimal" power ratios, backing ring dimensions and growth recipes into constantly moving targets throughout the campaign. One of our present manufacturing improvement initiatives is to develop an efficient means of treating a freshly cleaned platen in such a way that it will approximate the emissivity of a well-used platen. In the meantime, we have developed an automated means of responding to platen-to-platen variation in the thermocouple setpoint needed to hit a wafer temperature target. We are also developing an automated means of identifying outlier platens and responding to the gradual drift in the optimum power ratio setting.

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